

Chiral Symmetry and Functional Integral

R. E. GAMBOA SARAVÍ, M. A. MUSCHIETTI, F. A. SCHAPOSNIK, AND J. E. SOLOMIN

*Departamento de Física, Facultad de Ciencias Exactas, Universidad Nacional de La Plata.
Comisión de Investigaciones Científicas de la Provincia de Buenos Aires, Argentina*

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The change in the fermionic functional integral measure under chiral rotations is analyzed. Using the ζ -function method, the evaluation of chiral Jacobians to theories including *non-hermitian* Dirac operators D , can be extended in a natural way. (This being of interest, for example, in connection with the Weinberg–Salam model or with the relativistic string theory.) Results are compared with those obtained following other approaches, the possible discrepancies are analyzed and the equivalence of the different methods under certain conditions on D is proved. Also shown is how to compute the Jacobian for the case of a *finite* chiral transformation and this result is used to develop a sort of path-integral version of bosonization in $d = 2$ space-time dimensions. This result is used to solve in a very simple and economical way relevant $d = 2$ fermionic models. Furthermore, some interesting features in connection with the θ -vacuum in $d = 2, 4$ gauge theories are discussed.

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1. INTRODUCTION

Anomalous Ward identities related to chiral symmetry were first studied by Adler [1] and Bell and Jackiw [2] in connection with the neutral pion decay [3], some fifteen years ago. Since then, chiral anomalies have been playing a role of increasing importance in particle physics. In particular, after the works of Gross and Jackiw [4], Georgi and Glashow [5] and Bouchiat, J. Iliopoulos and Meyer [6] their importance in the renormalization of gauge theories became apparent.

More recently, after the work of 't Hooft [7] on symmetry breaking through chiral anomalies, their mathematical significance was clarified. Following 't Hooft work, and those of Jackiw and Rebbi [8] and Callan, Dashen and Gross [9], in which the instanton contributions were studied, the relation between anomalies and the index theorem was stressed and many important features of this mathematical aspect were analysed (see, for example, Refs. [10–12]).

The relevance of anomalies in composite models—'t Hooft matching conditions [13], in QCD effective Lagrangians [14] and in monopole catalysis of proton decay [15] have been discovered recently. The subject is also presently studied in connection with supersymmetry [16] as well as with physical theories in higher dimensional spaces [17].

The original works on chiral anomalies were based on perturbative analysis. In

their paper on chiral models, Gross and Jackiw [4] used the path-integral framework, which seems to be the most adequate to evade the diagrammatic analysis.

Fujikawa [18] was the first to develop a methodology which allows the study of chiral anomalies in the path-integral framework, independently of perturbation theory. In his work he made the important observation that the path-integral measure, the only place where the quantum aspect appears in this approach, is not invariant under chiral rotations. He was able to show that the associated Jacobian was responsible for the anomalous term in the corresponding Ward–Takahashi identities. He then extended his method to treat more general gauge theories and also derived the trace anomaly of the energy-momentum tensor in a very simple and elegant way [19].

His ideas were also employed to implement a sort of path-integral version of the bosonization technique in two dimensional models. Very simple solutions of the Schwinger and Thirring models were obtained in this way [20, 21] but also new insights on other not completely solved yet models were got, as it is the case of QCD_2 , the chiral Gross–Neveu and CP^n models [21–23].

Very recently, when Fujikawa's approach was used to study models where fermions were coupled to axial vectors or pseudoscalars, discrepancies with certain perturbative results were discovered [24–27]. This cast some doubts on the method validity when applied to theories which included γ_5 -couplings and originated a discussion on the way the regularization of the Jacobian has to be performed.

It is the purpose of this work to discuss this last problem using the ζ -function method, introduced in this context by Hawking [28] and already used for the computation of chiral Jacobians in Refs. [29, 30].

In this approach, the quadratic fermionic path-integral is regularized from the beginning using the ζ -function

$$\int \bar{\psi} \psi \exp \left(- \int \bar{\psi} D \psi dx \right) \equiv \exp \left\{ - \frac{d\zeta}{ds}(s, D) \right\}_{s=0} \quad (1.1)$$

The novelty of the approach we shall present is that we are able to work directly with the operator D (which shall be called, in general, the Dirac operator) without necessity of using the hermitian operator $D^+ D$ as it was done in the precedent works [29, 30]. As we shall see, this will allow us to treat non-hermitian, and even non-normal (i.e., $DD^+ \neq D^+ D$) operators, containing for example γ_5 -couplings on an equal foot with the hermitian ones.

From Eq. (1.1) one can compute the Jacobian arising from any fermion transformation $\Omega, \tilde{\Omega}$

$$\begin{aligned} \psi &= \Omega \psi' \\ \bar{\psi} &= \bar{\psi}' \tilde{\Omega} \end{aligned} \quad (1.2)$$

just as the ratio of two (already regularized) determinants

$$\log J = \zeta'(0, \tilde{\Omega} D \Omega) - \zeta'(0, D). \quad (1.3)$$

Being the r.h.s. in Eq. (1.3) finite, no regularization for the Jacobian is needed and hence, once definition (1.1) is established, no alternative between different Jacobian regularizations appears.

We shall see that for the case of hermitian D 's, our approach is equivalent to the other ones. But this is no more true for the case of non-hermitian Dirac operators: the ζ -function method coincides with some regularization techniques and disagrees with others.

The outline of the paper is as follows: We describe in Section 2 how the ζ -function can be employed in the computation of the change in the path integral measure under chiral rotations. Our method only requires D to be elliptic and invertible and hence it is particularly apt for the cases in which D is non-hermitian (interesting, for example, in the context of the Weinberg–Salam theory). As an example we consider $D = i\partial + \gamma_5 \mathcal{A}$, a non-hermitian operator, and compute the chiral Jacobian. A discussion of the extension of our technique to non-invertible operators is also given.

We then review in Section 3 Fujikawa's approach to the computation of chiral Jacobians, putting special emphasis on the modifications which have to be performed in order to treat, within this framework, theories which contain γ_5 -couplings.

The equivalence between Fujikawa's approach and the ζ -function method is proved in Section 4 for hermitian operators. This proof suggests the use of the complete Dirac operator (the one appearing in the fermionic Lagrangian) and not just a gauge-invariant part of it in the regularization of the Jacobian.

We discuss in Section 5 the application of the method we developed to the study of some field-theoretical models. In particular we show how finite chiral transformations have to be handled and then solve several interesting 2-dimensional fermionic models. We also discuss some implications of this approach to the computation of vacuum expectation values in gauge theories when the θ -vacuum is considered.

We give in Section 6 a summary of our results and conclusions and show in the Appendix how Seeley coefficients, appearing in the ζ -function approach are computed.

2. THE ζ -FUNCTION APPROACH

We shall explain in this section how the ζ -function method can be used to give meaning to a quadratic fermionic path integral and then show how the change in the fermionic measure under infinitesimal transformations can be handled. In particular, the evaluation of the Jacobian associated with a chiral rotation, for a theory with a non-hermitian Dirac operator (containing an axial vector) is discussed as an example of the method.

In the computation of quadratic path-integrals one is led to the evaluation of determinants of differential operators. These determinants diverge because the product of eigenvalues $\prod_j \lambda_j$ increases without bound and it is therefore necessary to adopt a regularization procedure. A technique which has proved to be very useful is the ζ -

function regularization [28]. Given an elliptic invertible operator D of order $m > 0$ ¹ defined on a compact manifold without boundary, M , of dimension d , one forms a generalized ζ -function from the eigenvalues of D

$$\zeta(s, D) = \sum_j \lambda_j^{-s}. \quad (2.1)$$

This series converges only for $\text{Re } s > d/m$, but $\zeta(s, D)$ can be analytically extended to a meromorphic function of s in the whole complex plane [31]. In particular, ζ is regular at $s = 0$ and its derivative at $s = 0$ is formally equal to $-\sum_j \log \lambda_j$. One can then define the regularized determinant of D , $\det D$, to be $\exp\{- (d\zeta/ds)(s, D)\} |_{s=0}$.

Actually Eq. (2.1) defines the ζ -function only if D is normal. If one does not assume normality, the definition of the ζ -function can be extended in the following way:

$$\zeta(s, D) = \int d\mu \, \text{tr } K_{-s}(D; x, x).$$

Here tr means the matricial trace and $d\mu$ is the measure in M . $K_{-s}(D; x, x)$ is the meromorphic extension of the evaluation of the kernel of D^{-s} on the diagonal $x = y$ (D^{-s} has a continuous kernel for $\text{Re } s$ sufficiently large [31]).

One can now define the regularized (Euclidean) quadratic path-integral

$$\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int \bar{\psi} D \psi \, dx \right) = \det D \equiv \exp(-\zeta'(0, D)). \quad (2.2)$$

We shall consider field theories in d -dimensional space-time and assume that all fields behave at infinity so that it is possible to compactify $\mathbb{R}^d$, for example to S^d by stereographic projection. Hence the elliptic operator D acts on the sections of a vector fiber bundle of fermions $\bar{\psi}, \psi$ over the compact manifold M . In many cases of interest D is the covariant derivative associated with the gauge group G , but it can also have a general form, including axial or pseudoscalar terms and in some cases it can be non-hermitian and even non-normal.

Expression (2.2) yields to a natural definition for the Jacobian associated to a fermion transformation. In particular, if we perform the infinitesimal chiral transformation

$$\begin{aligned} \psi &= e^{\gamma_5 \delta \alpha(x)} \psi', \\ \bar{\psi} &= \bar{\psi}' e^{\gamma_5 \delta \alpha(x)}, \end{aligned} \quad (2.3)$$

in the path-integral (2.2) we obtain the relation

$$\det D = J \det(e^{\gamma_5 \delta \alpha(x)} D e^{\gamma_5 \delta \alpha(x)}), \quad (2.4)$$

¹ In fact, one also has to assume that the principal symbol of D satisfies Seeley's hypothesis [31] which are sufficiently general as to be fulfilled by all interesting D .

where J is the Jacobian associated with the chiral change (2.3). As we stressed in the Introduction, J , defined from (2.2) and (2.4) is finite since it is the ratio of two regularized determinants. On the contrary, in Fujikawa's approach it is precisely on J that the regularization has to be carried out and many alternative choices appear.

Now, the operator $e^{\gamma_5 \delta \alpha} D e^{\gamma_5 \delta \alpha}$ can be rewritten as

$$e^{\gamma_5 \delta \alpha(x)} D e^{\gamma_5 \delta \alpha(x)} = D + [\gamma_5 \delta \alpha D + D \gamma_5 \delta \alpha] + O(\delta \alpha^2). \quad (2.5)$$

We shall assume that the principal part of D anticommutes with γ_5^2 . This is the case for all D we are interested in (whose principal part is $i\not{\partial}$). With this hypothesis the perturbations of D are of order strictly less than m . Under this condition the results of Ref. [30] are valid. From (2.2) and (2.5) we have

$$\log J = \zeta'(0, D + A_1) - \zeta'(0, D), \quad (2.6)$$

where

$$A_1 = \gamma_5 \delta \alpha D + D \gamma_5 \delta \alpha. \quad (2.7)$$

From the ζ -function differentiability results [30] we have

$$\log J = -\frac{d}{ds} [s \operatorname{Tr}(D^{-s-1} A_1)]. \quad (2.8)$$

Here $\operatorname{Tr}(D^{-s-1} A_1)$ corresponds to the usual operator trace, defined only for sufficiently large s . In a neighborhood of $s=0$ we take the analytic extension as considered in Ref. [31]. Now, for appropriate s , D^{-s} is a trace-class operator. So, for such s , $D^{-s-1} A_1$ is also a trace class operator, since it is the composition of D^{-s} with the bounded operator $(D^{-1} \gamma_5 \delta \alpha D + \gamma_5 \delta \alpha)$. Then, we use the cyclic properties of the trace and, extending analytically to $s=0$, we obtain

$$\log J = -2 \frac{d}{ds} [s \operatorname{Tr}(D^{-s} \gamma_5 \delta \alpha)]. \quad (2.9)$$

According to Seeley [31] the trace in (2.9) can be written as

$$\operatorname{Tr}(D^{-s} \gamma_5 \delta \alpha) = \int_{S^d} d\mu \operatorname{tr}(K_{-s}(D; x, x) \gamma_5) \delta \alpha. \quad (2.10)$$

The Kernel K_{-s} is regular at $s=0$ and can be explicitly evaluated by means of Seeley's coefficients b_p [31].

² We work in a representation in which the γ_μ -matrices are hermitian and satisfy $\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = \delta_{\mu\nu}$, $\gamma_5 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$.

The Jacobian associated with transformation (2.3) is hence given by

$$J = \exp \left\{ -2 \int_{S^d} d\mu \delta\alpha(x) \operatorname{tr}(K_0(D; x, x) \gamma_5) \right\}. \quad (2.11)$$

If D is hermitian expression (2.11) can also be written as

$$J = \exp \left\{ -2 \int_{S^d} d\mu \delta\alpha(x) \operatorname{tr}(K_0(D^2; x, x) \gamma_5) \right\}, \quad (2.12)$$

since, in this case, $K_0(D; x, x) = K_0(D^2; x, x)$ as we shall prove in the proposition at the end of this section. Expression (2.12) for J will be used to establish the equivalence between Fujikawa's approach and the ζ -function method. But expression (2.11) is more general than (2.12) since it applies to any elliptic operator, not necessary hermitian.

We shall use Eq. (2.11) in order to compute the jacobian for a theory in $d=4$ (Euclidean) dimensions, with a non-hermitian D given by

$$D = i\cancel{D} + \gamma_5 \mathcal{A}. \quad (2.13)$$

For simplicity we take $\mathcal{A}_\mu$ as the potential of an Abelian gauge theory.

Before proceeding to the evaluation of J , we mention that for the case of the ordinary operator, $D = i\cancel{D} + \mathcal{A}$, the computation of (2.11) or (2.12) leads to the well-known anomaly result, as was shown in Ref. [30].

Now, for operator (2.13), in order to compute $K_0(D; x, x)$ we follow Seeley technique [31]. The symbol of the differential operator D is

$$\sigma(D) = a_1 + a_0, \quad (2.14)$$

where a_1 is the principal symbol of D ,

$$a_1 = -\cancel{D} \quad (2.15)$$

and a_0 is given by

$$a_0 = \gamma_5 \mathcal{A}.$$

We have to construct the coefficients $b_{-1-j}(x, \xi, \lambda)$, $j=0, \dots, d=4$, in order to use Seeley's result for $K_0(D; x, x)$

$$K_0(D; x, x) = \frac{-i}{(2\pi)^d} \int_{|\xi|=1} d\xi \int_0^\infty b_{-1-j}(x, \xi, it) dt. \quad (2.16)$$

The b_i 's satisfy relations

$$\begin{aligned} b_{-1}(a_1 - \lambda) &= I \\ b_{-1-l}(a_1 - \lambda) + \sum \left(\frac{\partial}{\partial \xi} \right)^\alpha b_{-1-j} \left(\frac{i\partial}{\partial x} \right)^\alpha a_{1-k} \frac{1}{\alpha!} &= 0, \quad l > 0, \end{aligned} \quad (2.17)$$

with the sum taken for $j < l$ and $j + k + |\alpha| = l$.

We show in the Appendix how the coefficients are computed. Using this result and Eq. (2.16) we get

$$\begin{aligned} \text{tr}(\gamma^5 K_0(D; x, x)) &= \text{tr} \left\{ -\frac{1}{3} \frac{1}{16\pi^2} {}^*F_{\mu\nu} F_{\mu\nu} \right. \\ &\quad \left. - \frac{i}{8\pi^2} \frac{1}{3} \partial_\mu [-4A^2 A_\mu + \partial^2 A_\mu] \right\}. \end{aligned} \quad (2.18)$$

Introducing this expression in Eq. (2.11) we finally get the Jacobian associated with the chiral infinitesimal rotation (2.3)

$$\begin{aligned} J &= \exp \left[2 \int d^4x \, \delta\alpha(x) \text{tr} \left(\frac{1}{48\pi^2} {}^*F_{\mu\nu} F_{\mu\nu} \right. \right. \\ &\quad \left. \left. + \frac{i}{24\pi^2} \partial_\mu (-4A^2 A_\mu + \partial^2 A_\mu) \right) \right]. \end{aligned} \quad (2.19)$$

The second term in the integrand in this last expression can be cancelled in the path-integral by adding a counterterm $\Delta\mathcal{L}$ to the original Lagrangian density. This method has been suggested by Bardeen [42] and used in this context in Refs. [25, 26, 29]. After the subtraction of this term, which does not contribute to the axial Ward–Takahashi identity, the “minimal Jacobian” takes the form

$$J_{\text{minimal}} = \exp \left[2 \int d^4x \, \left(\frac{1}{48\pi^2} \text{tr} {}^*F_{\mu\nu} F_{\mu\nu} \right) \delta\alpha(x) \right]. \quad (2.20)$$

Throughout this section, the condition of invertibility was imposed on D . It is well-known, however, that in many cases of interest the operator D admits $\lambda = 0$ as eigenvalue. Any natural definition of $\det D$ should vanish in this case. However, it is of interest to keep the “product” of the non-zero eigenvalues, $\det' D$. This can be done for hermitian D 's by introducing the definition [30]

$$\det' D = \lim_{\alpha \rightarrow 0^+} \frac{\det(D + \alpha I)}{\alpha^N}, \quad (2.21)$$

where N is the dimension of the null set of D .

It has been proved in Ref. [30] that $\det' D = \det(D + P_{\text{ker}})$, where P_{ker} is the projection onto the null set of D .

From definition (2.21), a similar treatment to that leading to expression (2.11) could be carried out for non-hermitian operators without difficulties.

The ζ -function method has been used in the literature in a different way. One can define the regularized determinant of an operator D as

$$\det D = e^{-\zeta'(0, D^+ D)^{1/2}}. \quad (2.22)$$

Note that $D^+ D$ is an hermitian semi-positive operator for every D .

For an hermitian D Eq. (2.22) reads

$$\det D = e^{-\zeta(0, D^2)^{1/2}}. \quad (2.23)$$

In this framework the Jacobian arising from the chiral transformation (2.3) is [30]

$$\log J = -\frac{1}{2} \frac{d}{ds} [s \operatorname{Tr}((D^2)^{-s-1} A'_1)], \quad (2.24)$$

where

$$A'_1 = \gamma_5 \delta\alpha D^2 + 2D\gamma_5 \delta\alpha D + D^2\gamma_5 \delta\alpha. \quad (2.25)$$

As above, we can use the cyclic properties of the trace and write

$$J = \exp \left\{ -2 \int d\mu \delta\alpha(x) \operatorname{tr}(K_0(D^2; x, x) \gamma_5) \right\}$$

which is (2.12).

We are going to end this section establishing the equivalence between Eq. (2.11) and (2.12). Under the hypothesis assumed for D throughout this section, we have:

PROPOSITION. *If D is hermitian of odd order m and the dimension d of M is even, then*

$$K_0(D; x, x) = K_0(D^2; x, x). \quad (2.26)$$

Proof. For $\operatorname{Re}(-s) > 0$ and sufficiently large,

$$\begin{aligned} K_s(D; x, x) &= g_s(x) + (-1)^s h_s(x) \\ K_s(-D; x, x) &= (-1)^s g_s(x) + h_s(x) \\ K_{s/2}(D^2; x, x) &= g_s(x) + h_s(x), \end{aligned} \quad (2.27)$$

where, for $D\varphi_n = \lambda_n \varphi_n$,

$$g_s(x) = \sum_{\lambda_n > 0} \lambda_n^s \varphi_n(x) \otimes \varphi_n^+(x)$$

and

$$h_s(x) = \sum_{\lambda_n < 0} |\lambda_n|^s \varphi_n(x) \otimes \varphi_n^+(x).$$

Since $K_{s/2}(D^2; x, x)$ and $K_s(D; x, x)$ can be analytically extended to $s = 0$, and

$$((-1)^s - 1) h_s(x) = K_s(D; x, x) - K_{s/2}(D^2; x, x),$$

$h_s(x)$ admits a meromorphic extension, having at the most a single pole at $s = 0$, from Eq. (2.27) so does $g_s(x)$.

It is easy to see from the definition of the Seeley's coefficient b_{-m-d} , Eq. (2.17) and from its homogeneity [31], that

$$\begin{aligned} b_{-m-d}(-D, x, \xi, -it) &= b_{-m-d}(D, x, \xi, it) \\ &= (-1)^{-(m+d)-1} b_{-m-d}(D, x, -\xi, it). \end{aligned}$$

Now, by using Eq. (2.16) we get

$$K_0(-D; x, x) = K_0(D; x, x)$$

and then, since

$$K_0(D; x, x) + K_0(-D; x, x) = 2K_0(D^2; x, x)$$

we have $K_0(D; x, x) = K_0(D^2; x, x)$.

3. FUJIKAWA'S APPROACH TO THE EVALUATION OF CHIRAL JACOBIANS

We begin this section by recapitulating Fujikawa's original approach to the evaluation of chiral Jacobians and then discuss the modifications which have to be introduced when operators others than the (hermitian) Dirac one appear in the fermionic Lagrangian.

The modifications, which were not necessary in the approach discussed in the preceding section, amount to a careful choice of a *regularization operator* (R.O.) which has to be used in order to make the Jacobian finite. This choice has to be performed according to the following prescription:

- (i) The selected R.O. has to be related to the one appearing in the fermionic Lagrangian.
- (ii) For the case a non-hermitian operator appears in the Lagrangian, the need for a positive semi-definite R.O. has to be taken into account.

For hermitian operators, Fujikawa's approach coincides with the ζ -function one and gives the already known results from perturbative calculations *provided the R.O.*

in Fujikawa's approach is taken as the square of the complete Dirac operator appearing in the Lagrangian. We have recently shown that this choice is justified by the ζ -function method, in which no alternative between complete or incomplete R.O. is present [32].

For the case of non-hermitian operators, different R.O. have been used in the literature [24–27, 29]. In particular, the analytic extension of a non-hermitian D to an hermitian form, and its use as R.O. in the regularization of the Jacobian agrees with the result obtained in the preceding section but disagrees with that, closer to Fujikawa's spirit [27] which should indicate the use of D^+D as R.O.

We leave for the next section the discussion of the equivalence between different approaches and describe now Fujikawa's method.

As we stressed before, it starts from the observation that the fermionic path-integral measure is not invariant under all relevant field transformations. The resulting Jacobian arising, for example, from a chiral one, has to be regularized, and Fujikawa's prescription is chosen gauge-invariant, this rendering non-trivial the chiral Jacobian, for a gauge theory coupled with fermions.

Consider, for example transformation (2.3) for the case of a $U(1)$ chiral rotation. In the path-integral framework, the fermionic measure is defined by expanding the (classical) fermion fields in terms of some complete set of functions φ_n, φ_n^+ :

$$\psi(x) = \sum_n a_n \varphi_n(x) \quad (3.1)$$

$$\bar{\psi}(x) = \sum_n \bar{a}_n \varphi_n^+(x), \quad (3.2)$$

where a_n and $\bar{a}_n$ are elements of an infinite Grassmann algebra. The fermionic measure is then defined as

$$\mathcal{L} \bar{\psi} \mathcal{L} \psi = \prod_n d\bar{a}_n da_n. \quad (3.3)$$

In terms of the set $\{\bar{a}_n, a_n\}$ transformation (2.3) can be written as

$$\begin{aligned} a &= \mathbb{C} a' \\ \bar{a} &= \bar{a}' \mathbb{C}, \end{aligned} \quad (3.4)$$

where

$$\begin{aligned} \mathbb{C}_{mn} &= \int \varphi_m^+(x) e^{i\gamma_5 \delta\alpha(x)} \varphi_n(x) dx \\ &\simeq \delta_{mn} + \int \varphi_m^+(x) \gamma_5 \varphi_n(x) \delta\alpha(x) dx. \end{aligned} \quad (3.5)$$

Hence, the Jacobian associated with transformation (2.3) (or equivalently (3.4)) is defined here as

$$\mathcal{L} \bar{\psi} \mathcal{L} \psi = J \mathcal{L} \bar{\psi}' \mathcal{L} \psi'. \quad (3.6)$$

Using Eqs. (3.5) and (3.6) we get

$$J = \exp \left[-2 \int dx A(x) \delta \alpha(x) \right], \quad (3.7)$$

where

$$A(x) = \sum_n \varphi_n^+(x) \gamma_5 \varphi_n(x). \quad (3.8)$$

It is important to note that expression (3.8) is only formal and it has no meaning until a regularization prescription is introduced [10]; Fujikawa's regularization choice, for a theory with fermionic Lagrangian

$$\mathcal{L} = \bar{\psi}(i\cancel{D} + eA) \psi = \bar{\psi} \cancel{D} \psi, \quad (3.9)$$

is based on the introduction of a cut-off t in the form

$$A_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \sum_n \varphi_n^+(x) \gamma_5 e^{-\lambda_n^2 t} \varphi_n(x), \quad (3.10)$$

where λ_n are the eigenvalues of the (hermitian) operator $\cancel{D}$ whose eigenfunctions

$$\cancel{D} \varphi_n(x) = \lambda_n \varphi_n \quad (3.11)$$

are chosen as the basis set in expansion (3.1)–(3.2).

Of course the resulting Jacobian cannot depend on the basis choice since the fermionic measure has to be defined independently of the complete set one selects. On the contrary, J does depend on the regularization prescription. For example, regularization (3.10) which can be written for the present basis as

$$A_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \sum_n \varphi_n^+(x) \gamma_5 e^{-\cancel{D}^2 t} \varphi_n(x) \quad (3.12)$$

is manifestly gauge-invariant. On the other hand, there are other possible cutoffs as, for example,

$$\tilde{A}_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \sum_n \varphi_n^+(x) \gamma_5 e^{-\cancel{D} t} \varphi_n(x) \quad (3.13)$$

which obviously spoil gauge invariance, even if the φ_n 's are chosen as eigenfunctions of $\cancel{D}$ [33]. In fact, $\tilde{A}_{\text{reg}} = 0$ and chiral symmetry is in this case maintained. This is a manifestation, in the path-integral framework, of the well-known fact that it is not possible to maintain both gauge and chiral symmetry.

In summary, $A_{\text{reg}}(x)$ is independent of the basis choice but depends on the selected regulator. For the gauge invariant R.O., $\exp(-t\mathcal{D}^2)$ the result is [19, 20]

$$A_{\text{reg}}(x) = -\frac{e^2}{16\pi^2} \text{tr} {}^*F_{\mu\nu} F_{\mu\nu} \equiv \rho_4(x) \quad \text{for } \begin{matrix} d=4, \\ SU(N) \text{ theory,} \end{matrix} \quad (3.14)$$

$$A_{\text{reg}}(x) = -\frac{e}{4\pi} \varepsilon_{\mu\nu} F_{\mu\nu} \equiv \rho_2(x) \quad \text{for } \begin{matrix} d=2 \\ U(1) \text{ theory} \end{matrix} \quad (3.15)$$

The Jacobian is then

$$\begin{aligned} J &= \exp \left\{ -2 \int A_{\text{reg}}(x) \delta\alpha(x) d^d x \right\} \\ &= \exp \left\{ -2 \int \rho_d(x) \delta\alpha(x) d^d x \right\}. \end{aligned} \quad (3.16)$$

As explained by Fujikawa, the anomalous Ward–Takahashi identities can be obtained by making the change of variables (2.3) in the generating functional Z of the theory and then using the fact that $\delta Z / \delta\alpha(x) = 0$. The result coincides with the usual perturbative one:

$$\begin{aligned} \partial_\mu J_\mu^5 &= \rho_d(x) \\ j_\mu^5(x) &= \bar{\psi} \gamma_\mu \gamma_5 \psi. \end{aligned} \quad (3.17)$$

It is just the Jacobian (3.16) the responsible for the r.h.s. in (3.17). As stated in Section 2 the same result can be obtained using the ζ -function method [29, 30] and this is not a fortuitous coincidence. There is a close relation between the ζ -function approach and Fujikawa's method, for the case of hermitian operators, as we shall see in the next section.

One could think that the reason why both methods coincide (and disagree with the trivial result derived from (3.14)) is that both of them respect gauge-invariance. However, as pointed out in Refs. [24, 33], the preservation of gauge symmetry is not the only criterion to be followed in order to correctly compute the Jacobian associated with chiral rotations. This was first discussed when Fujikawa's ideas were applied to solve the Schwinger model [20] ($d=2$, $U(1)$ gauge theory with massless fermions) in its path-integral formulation. In this approach one has to consider a “decoupling change” in the field variables

$$\begin{aligned} \psi &= e^{\gamma_5 \alpha(x)} \psi' \\ \bar{\psi} &= \bar{\psi}' e^{\gamma_5 \alpha(x)}, \end{aligned} \quad (3.18)$$

$$A_\mu = -\frac{1}{e} \varepsilon_{\mu\nu} \partial_\nu \alpha(x), \quad (3.19)$$

and is then faced to a *finite* chiral rotation, Eq. (3.18). As it was stressed in Ref. [20], at each stage in the iteration of the infinitesimal rotations needed to build up the full transformation (3.18) the Lagrangian is modified: If we write for the chiral rotations

$$\begin{aligned}\psi &= e^{\gamma_5 \alpha(x)r} \psi'_r \\ \bar{\psi} &= \bar{\psi}'_r e^{\gamma_5 \alpha(x)r},\end{aligned}\quad (3.20)$$

with $r \in (0, 1)$,

$$\mathcal{L}' = \bar{\psi}'_r (i\partial\!\!\!/ + e\mathcal{A}\!\!\!/ + ir\gamma_\mu\gamma_5\partial_\mu\alpha(x)) \psi'_r. \quad (3.21)$$

We then see that in the interval $(0, 1)$ the Dirac operator now reads

$$D_r = \mathcal{D} - ir\gamma_5\partial\!\!\!/ \alpha = i\partial\!\!\!/ + e(1-r)\mathcal{A}\!\!\!/. \quad (3.22)$$

Both $\mathcal{D}$ and D_r are gauge covariant and in principle can be used to regulate $A(x)$ à la Fujikawa. It is, however, D_r that has to be employed in order to get the correct “photon” mass from the Jacobian. (See Ref. [20] for the details within this approach and Section 5 in this paper for the discussion of this model in the ζ -function approach.) The final result is that the Jacobian, $J = \exp(-(e^2/2\pi) \int A^\mu A_\mu d^2x)$ gives the well-known photon mass term of the Schwinger model.

This example, as well as those provided by the analysis of the QCD_2 , Thirring, chiral Gross–Neveu and CP^n models [21–23] indicate that for *finite* chiral rotations the R.O. selected to evaluate $A_{\text{reg}}(x)$ à la Fujikawa has not to be just $\mathcal{D}$ but the complete operator appearing at each stage of the iteration.

More recently the investigation of theories including scalar, pseudoscalar, vector and axial sources [24–27] has also shown that in following Fujikawa’s approach a similar criterion to the one described above has to be followed.

Consider, for example, a theory with (Euclidean) fermionic Lagrangian

$$\mathcal{L} = \bar{\psi} D \psi, \quad (3.23)$$

D being the elliptic hermitian operator

$$D = i\partial\!\!\!/ + \not{V} + i\gamma_5 \not{A} + j \frac{(1 + \gamma_5)}{2} + k \frac{(1 - \gamma_5)}{2} \quad (3.24)$$

containing vector (V_μ), axial (A_μ), scalar ($j + k$) and pseudoscalar ($j - k$) couplings. Here,

$$V_\mu = V_\mu^a t^a, \quad A_\mu = A_\mu^a t^a, \quad (3.25)$$

where t^a are the hermitian generators of G in some representation. j and k can be also taken in the form

$$j = j^a t^a, \quad k = k^a t^a \quad (3.26)$$

although in the example we shall discuss we consider for simplicity that j and k are scalars. The fermions are assigned to the fundamental representation of G and can be made massive by adding a term to D :

$$D \rightarrow D + m. \quad (3.27)$$

As it has been stressed in Refs. [24–26, 32], in order to obtain the results for the anomaly reported in the literature (based on perturbative calculations), one must use, in the approach described in this section, the recipe

- (i) The complete operator D has to be used in the regularization of $A(x)$

$$A_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \sum_n \varphi_n^\dagger(x) \gamma_5 e^{-tm^2} \varphi_n(x) \quad (3.28)$$

$$D\varphi_n = \lambda_n \varphi_n.$$

- (ii) All residual (possible divergent) terms appearing in A_{reg} have to be removed.

As an example of the application of this recipe, let us consider the simple case in which $A_\mu = 0$ and j and k are pure scalars. Use of Eq. (3.28) leads to

$$A_{\text{reg}}^d(x) = \lim_{t \rightarrow 0^+} \text{tr} \left\{ \gamma_5 \int \frac{d^d p}{(2\pi)^d} e^{-ipx} e^{-D^2 t} e^{ipx} \right\}$$

$$= \lim_{t \rightarrow 0^+} \text{tr} \left\{ \gamma_5 \int \frac{d^d p}{(2\pi)^d} e^{-(D^2 - p^2)t} \right\}, \quad d = 2, 4. \quad (3.29)$$

Expanding the exponential in (3.29) one gets for $d = 4$

$$A_{\text{reg}}^4(x) = \lim_{t \rightarrow 0^+} \left\{ -\frac{1}{8\pi^2} \frac{1}{t} (j^2 - k^2) + \frac{1}{16\pi^2} \left[-\text{tr} {}^*F_{\mu\nu} F_{\mu\nu} \right. \right.$$

$$+ \frac{1}{3} (j^4 - k^4) - \frac{4}{3} jk(j^2 - k^2) + \frac{4}{3} (k\partial^2 k - j\partial^2 j)$$

$$+ \left. \frac{2}{3} (j\partial^2 k - k\partial^2 j) + (\partial_\mu k)^2 - (\partial_\mu j)^2 \right]$$

$$+ O(t) \Big\}. \quad (3.30)$$

Note that a residual divergent term is still present in A_{reg}^4 . There are several subtraction schemes, based on Bardeen ideas [42], which can be used in order to get the final finite and minimal result (we have already mentioned this point in Section 2). One then gets

$$\begin{aligned}
A_{\text{fin}}^4(x) = & \rho_4 + \frac{1}{16\pi^2} \left[\frac{1}{3} (j^4 - k^4) - \frac{4}{3} jk(j^2 - k^2) \right. \\
& - \frac{4}{3} (j\partial^2 j - k\partial^2 k) + \frac{3}{3} (j\partial^2 k - k\partial^2 j) \\
& \left. - (\partial_\mu j)^2 + (\partial_\mu k)^2 \right], \tag{3.31a}
\end{aligned}$$

$$A_{\text{reg}}^2(x) = \rho_2 - \frac{1}{4\pi} (j^2 - k^2), \tag{3.31b}$$

where we have also given the two-dimensional result for future task. (There is no need in the $d = 2$ case of further subtractions.)

Up to now we have explained the way one can compute the chiral Jacobian when an hermitian operator D appears in the Lagrangian. One can easily extend this prescription to normal operators. Formula (3.16) remains valid, if one writes for

$$A_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \sum \varphi_n^+(x) \gamma_5 e^{-D^+ D t} \varphi_n(x) \tag{3.22}$$

and

$$J = \exp \left\{ -2 \int A_{\text{fin}}(x) \delta\alpha(x) d^d x \right\}. \tag{3.33}$$

Again, one has to take the finite part of A_{reg} , subtracting the residual (possibly divergent) terms.

There are, however, cases of interest where the operator D appearing in the Lagrangian is non-hermitian and non-normal. This does not pose a problem in the ζ -function approach described in Section 2, where the only condition imposed on D was ellipticity. On the contrary, in the approach we are following in this section, it is evident that one needs a positive semi-definite R.O. defined from D and this makes the problem more involved.

We shall show how Fujikawa [27] extended his method to include this kind of operators. We shall then compare the extension with other possible ones. Consider for simplicity the case in which the non-hermitian Dirac operator is simply the one given in Eq. (2.13)

$$D = i\not{\partial} + \gamma_5 \not{A}$$

and define, following Fujikawa [27], the two positive semi-definite hermitian operators

$$\begin{aligned}
H_\psi &= D^+ D \\
H_{\bar{\psi}} &= D D^+. \tag{3.34}
\end{aligned}$$

One disposes now of two orthogonormal complete basis $\{\varphi_n\}$ and $\{\phi_n\}$

$$\begin{aligned} H_\psi \varphi_n &= \lambda_n^2 \varphi_n, & \lambda_n &\in \mathbb{R}, \\ H_{\bar{\psi}} \phi_n &= \lambda_n^2 \phi_n \end{aligned} \quad (3.35)$$

which shall be used in order to expand ψ and $\bar{\psi}$, respectively,

$$\begin{aligned} \psi &= \sum a_n \varphi_n \\ \bar{\psi} &= \sum \bar{a}_n \phi_n. \end{aligned} \quad (3.36)$$

With this choice, the fermionic action takes the form

$$S_F = \int \bar{\psi} D \psi \, dx = \sum_{\lambda_n \neq 0} \lambda_n \bar{a}_n a_n \quad (3.37)$$

due to the relation

$$\phi_n = \frac{1}{\lambda_n} D \varphi_n, \quad (3.38)$$

for $\lambda_n \neq 0$ ($D \varphi_n = 0$ for $\lambda_n = 0$). Although the λ_n in Eq. (3.35) are in general different from the eigenvalues of D , we see that in some sense one is “defining” the square root of $\det D^+ D$. Using $\exp(-tH)$ as regulator, Fujikawa has shown that under a chiral change, $A_{\text{reg}}(x)$ takes the form

$$A_{\text{reg}}(x) = \sum \varphi_n^+ \gamma_5 \varphi_n = -\frac{1}{16\pi^2} \text{tr} \, {}^* F_{\mu\nu} F_{\mu\nu}. \quad (3.39)$$

On the other hand, if one uses operator (2.13) analytically extended to an hermitian form, the result is [24–26]

$$A_{\text{fin}}(x) = -\frac{1}{3} \frac{1}{16\pi^2} \text{tr} \, {}^* F_{\mu\nu} F_{\mu\nu}. \quad (3.40)$$

We then see that Fujikawa’s approach leads to a result that does not coincide with the ζ -function method one (Eq. (2.20)), while the latter gives the same answer as those approaches where the non-hermitian operator is analytically extended to an hermitian form in order to construct the R.O. This can be understood if one notes that the Jacobian is a polynomial in A_μ (of degree ≤ 4) and hence the analytic extension of D to the hermitian operator

$$D_H = i\not{\partial} + i\gamma_5 \mathcal{A}$$

can be thought as the extension $A_\mu \rightarrow iA_\mu$ which is trivial for a polynomial. This explains the coincidence of the results found in Refs. [24, 26, 29] based on analytic

extensions with that of Section 2 where no distinction between hermitian or non-hermitian operators is done. On the other hand, expression (3.39) was obtained using D^+D as R.O., not related to the analytic extension of D and hence in this case there is no reason to expect any coincidence.

The ζ -function approach, as developed in Section 2, seems to be natural in the sense that it treats hermitian and non-hermitian operators on equal footing. Its discrepancy with the result obtained by using $\exp(-Ht)$ as R.O. is already present in the perturbative calculations and merits further discussion (see the references in the works [24–27]).

4. EQUIVALENCE BETWEEN ζ -FUNCTION AND FUJIKAWA'S METHODS

We prove in this section the equivalence between the ζ -function method described in section 2 and Fujikawa's approach, recapitulated in the preceding section.

We consider an elliptic invertible operator D of order $m > 0$ defined on a compact manifold M without boundary, of dimension d . In order to prove the equivalence proposition, we have to assume that D is hermitian. For such a class of operators it has been explained in Section 2 that the ζ -function method yields, for the infinitesimal transformation (2.3), to the Jacobian (Eq. (2.12))

$$J_\zeta = \exp \left\{ -2 \int \delta\alpha(x) \operatorname{tr} [K_0(D^2; x, x) \gamma_5] d^d x \right\}. \quad (4.1)$$

On the other hand, we have seen that Fujikawa's approach leads to the expression for J (Eq. (3.32)),

$$J_F = \exp \left\{ -2 \int \delta\alpha(x) A_{\text{fin}}(x) d^d x \right\}, \quad (4.2)$$

where $A_{\text{fin}}(x)$ means we have subtracted the eventual residual divergent terms from $A_{\text{reg}}(x)$:

$$A_{\text{fin}}(x) = \text{finite part} \left\{ \lim_{t \rightarrow 0^+} \sum \varphi_n^+(x) \gamma_5 e^{-tD^+D} \varphi_n(x) \right\}. \quad (4.3)$$

Here $\{\varphi_n\}$ is the orthonormal basis of eigenfunctions

$$D^+D\varphi_n = |\lambda_n|^2 \varphi_n. \quad (4.4)$$

Since D^+D is a self-adjoint positive elliptic operator, for $t > 0$, e^{-tD^+D} is an integral trace-class operator whose continuous Kernel $H(t, x, y)$ evaluated on the diagonal $x = y$ admits the asymptotic expansion

$$H(t, x, x) \sim \sum_{k=0}^{\infty} E_k(x) t^{(k-d)/2m} \quad \text{for } t \rightarrow 0^+. \quad (4.5)$$

(This expansion has been established in Refs. [34–35].) Now, from (4.3) and (4.5) we see that

$$A_{\text{reg}}(x) = \lim_{t \rightarrow 0^+} \text{tr}(H(t, x, x) \gamma_5). \quad (4.6)$$

We have explained that the finite part of $A_{\text{reg}}, A_{\text{fin}}$ can be obtained by an appropriate subtraction procedure. For the r.h.s. term in (4.6), its finite part is just

$$A_{\text{fin}}(x) = \text{tr}(E_d(x) \gamma_5). \quad (4.7)$$

Now, the equivalence between Eqs. (4.1) and (4.2) will follow from the next proposition which shall be proved by slight modifications of Seeley's arguments presented in Ref. [35].

PROPOSITION. *For an elliptic and invertible operator D the following identity holds:*

$$K_0(D^+ D; x, x) = E_d(x). \quad (4.8)$$

Proof. From the relation between $e^{-tD^+ D}$ and $(D^+ D)^s$ given by the Mellin transformation [35]

$$(D^+ D)^s = \frac{1}{\Gamma(-s)} \int_0^\infty dt e^{-tD^+ D} t^{-s-1} \quad \text{for } \text{Re } s < 0 \quad (4.9)$$

we have

$$K_s(D^+ D; x, x) = \frac{1}{\Gamma(-s)} \int_0^\infty t^{-s-1} H(t, x, x) dt. \quad (4.10)$$

This expression is valid for s such that $(D^+ D)^s$ has a continuous kernel (i.e., for $\text{Re } s < -d/2m$). In order to evaluate the analytic extension of $K_s(D^+ D; x, x)$ for $s = 0$ we shall first see that the integral in the r.h.s. of Eq. (4.10) can be extended to $s = 0$. From the behavior of $H(t, x, x)$ for $t \rightarrow \infty$ it can be shown that for any matricial norm $\| \cdot \|$, the estimate

$$\|H(t, x, x)\| \leq c(x) e^{(1-t)\varepsilon} \quad \text{for } t \rightarrow \infty \quad (4.11)$$

holds with $\varepsilon > 0$, $c(x) < \infty$ for almost every x . The proof of Eq. (4.11) starts from the expression

$$H(t, x, x) = \sum_n \varphi_n(x) \otimes \varphi_n^+(x) e^{-t|\lambda_n|^2} \quad (4.12)$$

by writing

$$\|H(t, x, x)\| \leq \sum_n \|\varphi_n(x) \otimes \varphi_n^+(x)\| e^{-|\lambda_n|^2} e^{(1-t)\varepsilon} \quad (4.13)$$

if $\varepsilon = \min (|\lambda_n|^2)$. Since

$$\|\varphi_n(x) \otimes \varphi_n^+(x)\| \leq \mathbb{C} \|\varphi_n(x)\|^2 \quad (4.14)$$

and

$$\int d^d x \sum_n \|\varphi_n(x)\|^2 e^{-|\lambda_n|^2} = \sum_n e^{-|\lambda_n|^2} = \text{Tr}(e^{-D+D}) < \infty \quad (4.15)$$

then

$$\sum_n \|\varphi_n(x) \otimes \varphi_n^+(x)\| e^{-|\lambda_n|^2} < \infty \quad \text{for almost every } x. \quad (4.16)$$

Hence the expression

$$\int_1^\infty t^{-s-1} H(t, x, x) dt \quad (4.17)$$

can be analytically extended to any $s \in \mathbb{C}$.

On the other hand for a sufficiently large r ,

$$\left\| H(t, x, x) - \sum_{k=0}^r E_k(x) t^{(k-d)/2m} \right\| \leq ct \quad (4.18)$$

and so

$$\int_0^1 t^{-s-1} \left\{ H(t, x, x) - \sum_{k=0}^r E_k(x) t^{(k-d)/2m} \right\} dt \quad (4.19)$$

can be analytically extended to $\text{Re } s < 1$.

Now from Eqs. (4.10) and (4.19) we obtain the expression for $K_{-s}(D^+D; x, x)$ in terms of the coefficients E_k ,

$$K_s(D^+D; x, x) = \frac{1}{\Gamma(-s)} \int_0^1 \sum_{k=0}^r E_k(x) t^{(k-d)/2m-s-1} dt + \frac{1}{\Gamma(-s)} h(s, x), \quad (4.20)$$

with $h(x, s)$ holomorphic for $\text{Re } s < 1$.

To wit

$$K_s(D^+D; x, x) = \frac{1}{\Gamma(-s)} \left[\frac{E_d(x)}{-s} + \sum_{k \neq d} E_k(x) \frac{1}{(k-d)/2m-s} + h(s, x) \right] \quad (4.21)$$

and hence the r.h.s. in Eq. (4.21) is a meromorphic function. Taking Eq. (4.21) for $s = 0$, we get (4.8).

Now, we insert (4.8) in (4.7) and then, from Eq. (4.6) we get

$$A_{\text{fin}}(x) = \text{tr}(K_0(D^+D; x, x) \gamma_5). \quad (4.22)$$

Using this result, we see that expression (4.2) coincides with (4.1) and hence the equivalence has been proved. So, for hermitian D 's, the regularized chiral Jacobian (4.2) obtained by using Fujikawa's prescription coincides with the one obtained by means of the ξ -function we proposed, (4.1). In this case, as we mentioned in Section 2, precisely the same Jacobian would have arisen if we had adopted from the beginning definition (2.22) as the regularized determinant of D .

Moreover, it can be shown, by slight modification of the arguments of Section 2, that for a normal (non-necessarily hermitian) operator D , the chiral Jacobian corresponding to definition (2.22) still agrees with the Fujikawa one [32].

5. APPLICATIONS

We shall apply in this section the techniques discussed above in order to study some interesting problems in quantum field theory. Since all these applications rest on the computation of Jacobians associated to finite chiral rotations, we shall begin by showing how the methods described above can be extended to handle this kind of transformations. We shall then use this result to develop a sort of path-integral version of the bosonization technique in 2-dimensional fermionic models and also discuss some interesting features of gauge theories when the θ -vacuum is considered.

In the preceding sections we have explained how the Jacobian corresponding to an infinitesimal chiral rotation can be computed. In order to extend the method to finite transformations

$$\begin{aligned}\psi &= e^{\gamma_5 \alpha(x)} \psi' \\ \bar{\psi} &= \bar{\psi}' e^{\gamma_5 \alpha(x)}\end{aligned}\tag{5.1}$$

and compute the associated Jacobian, we shall consider the generating functional

$$Z = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left\{ - \int \bar{\psi} D \psi d^d x \right\},\tag{5.2}$$

where D is the operator defined in Eq. (3.24). As we have seen in Section 2, Z can be regularized by means of the ξ -function method:

$$Z = \det D = \exp \left[- \frac{d\xi}{ds}(s, D) \right]_{s=0}.\tag{5.3}$$

Now, finite transformation (5.1) will be built up by successive infinitesimal changes. To this end, we introduce a real parameter r ($0 \leq r \leq 1$) and write

$$\begin{aligned}\psi'_r(x) &= e^{-\gamma_5 r \alpha(x)} \psi(x) \\ \bar{\psi}'_r(x) &= \bar{\psi}(x) e^{-\gamma_5 r \alpha(x)}.\end{aligned}\tag{5.4}$$

Starting from $r = 0$ we shall end up at $r = 1$ reaching the full transformation (5.2). If we explicitly perform the change of variables (5.4) in (5.3), we get

$$Z = \int J(r) \mathcal{D}\tilde{\psi}'_r \mathcal{D}\psi'_r \exp \left(- \int \tilde{\psi}'_r D_r \psi'_r d^d x \right), \quad (5.5)$$

where

$$D_r = e^{\gamma_5 r \alpha(x)} D e^{\gamma_5 r \alpha(x)} \quad (5.6)$$

and we have written

$$\mathcal{D}\tilde{\psi} \mathcal{D}\psi = J(r) \mathcal{D}\tilde{\psi}'_r \mathcal{D}\psi'_r. \quad (5.7)$$

The Jacobian we are searching for is $J = J(r)|_{r=1}$.

Now, for arbitrary r we can write

$$Z = J(r) \det D_r \quad (5.8)$$

and since Z cannot depend on r ,

$$\frac{dZ}{dr} = 0 = \frac{dJ(r)}{dr} \det D_r + J(r) \frac{d}{dr} (\det D_r). \quad (5.9)$$

By integrating (5.9) we finally get

$$J = J(1) = \exp \left[- \int_0^1 w'(r) dr \right], \quad (5.10)$$

where

$$w(r) = \log \det D_r. \quad (5.11)$$

We have then to compute $w(r)$ which is, by definition,

$$w(r) = - \frac{d}{ds} \zeta(s, D_r) |_{s=0}. \quad (5.12)$$

Now, exactly as in formulae (2.5)–(2.8) we have

$$D_{r+\Delta r} = D_r + A_1 \Delta r + O(\Delta r^2), \quad (5.13)$$

with

$$A_1 = \gamma_5 \alpha(x) D_r + D_r \gamma_5 \alpha(x). \quad (5.14)$$

Then,

$$w'(r) = \lim_{\Delta r \rightarrow 0} \frac{1}{\Delta r} (\log \det D_{r+\Delta r} - \log \det D_r). \quad (5.15)$$

Using Eqs. (5.14) and (5.12) and proceeding as in Section 2 we finally get

$$w'(r) = \lim_{\Delta r \rightarrow 0} \frac{-1}{\Delta r} [\zeta'(0, D_r + A_1 \Delta r) - \zeta'(0, D_r)] \\ = \frac{d}{ds} \left[s \operatorname{Tr}(D_r^{s-1} A_1) \right]_{s=0} = 2 \operatorname{Tr}(D_r^{-s} \gamma_s \alpha)|_{s=0},$$

or, from Eq. (5.10),

$$J = \exp \left[-2 \int_0^1 d^d x \int_0^1 dr \operatorname{Tr}(D_r^{-s} \gamma_s \alpha) \right]_{s=0}, \quad (5.16)$$

where the trace can be computed as explained in Section 2. Of course, for hermitian operators, the equivalence found in the preceding section allows us to write, instead of (5.16),

$$J = \lim_{t \rightarrow 0} \exp \left[-2 \int_0^1 d^d x \int_0^1 dr \operatorname{Tr}(e^{-t D_r^2} \gamma_s \alpha) \right], \quad (5.17)$$

but, as we explained above, Eq. (5.16) is far more general and has the advantage of giving the finite Jacobian without necessity of subtraction of residual (possibly divergent) terms.

2a. Two-Dimensional Models

Using the technique explained above one can develop a method which parallels in the path-integral framework the operator fit first given by Lowenstein and Swieca [36] for the Schwinger model and then extended to many two-dimensional models.

The method, which provides a sort of path-integral version of the bosonization technique, starts from the observation that in two dimensions a chiral change of the kind (5.1) decouples, at the classical level, fermions from the other fields present in many interesting theories. By studying the change in the fermionic measure, the only place where the quantum aspect plays a role in the path-integral, one can extend this decoupling to the quantum level. In such a way, a very simple and complete solution of models like massless QED_2 and the Thirring one can be obtained. Moreover, the method allows for new insights onto the not completely solved yet non-Abelian models (like QCD_2 , chiral Gross–Neveu or CP^n models). We shall briefly describe this method by considering a series of examples of increasing difficulty:

The Schwinger Model [20]

The dynamics of this Abelian model (for the original works, see references in [20]) is defined in (Euclidean) two-dimensional space-time by the Lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 + \bar{\psi}(i\partial + eA)\psi. \quad (5.18)$$

At the classical level, transformation (5.1) decouples the fermions from the gauge field provided one makes the choice

$$A_\mu = -\frac{1}{e} \varepsilon_{\mu\nu} \partial_\nu \alpha. \quad (5.19)$$

This is always possible in the Lorentz gauge, which we choose for simplicity. (The discussion can be extended without difficulty to an arbitrary gauge.)

After the change of variables (5.1) and (5.19), Lagrangian (5.18) becomes that of a free fermion decoupled from the gauge field

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 + \bar{\psi}' i \not{\partial} \psi'. \quad (5.20)$$

We note that the decoupling of fermions from the gauge field can be achieved even for A_μ 's belonging to a non-trivial topological sector. In this last case the chiral transformation (5.1) is topologically non-trivial and hence changes the fiber bundle in which D is defined. The computation of the Jacobian is then more involved and we do not have yet a rigorous derivation of the finite Jacobian, except for the zero topological-charge sector, where the decoupling transformation does not change the fiber bundle.

This restriction does not spoil, however, the physical information one can get in this approach: All Green functions and composite quantum matrix elements relevant for the computation of any structure function can be derived in complete agreement with the bosonization results not only for the Schwinger but also for the Thirring model (see Ref. [21]). Also, the fundamental properties of QCD_2 , chiral Goss-Neveu and CP^n models can be discovered in this approach.

We have to compute, in order to treat the problem at the quantum level, the Jacobian associated with transformation (5.1):

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi = J \mathcal{D}\bar{\psi}' \mathcal{D}\psi'. \quad (5.21)$$

We can use the machinery developed above. The operator D_r reads in this case

$$D_r = i\not{\partial} + e\not{A} + ir\gamma_\mu \gamma_5 \partial_\mu \alpha. \quad (5.22)$$

With our choice of γ -matrices, $-i\gamma_\mu \gamma_5 = \varepsilon_{\mu\nu} \gamma^\nu$ and hence

$$D_r = i\not{\partial} - (1-r) \gamma_\mu \varepsilon_{\mu\nu} \partial_\nu \alpha, \quad (5.23)$$

or

$$D_r = i\not{\partial} + e(1-r) \not{A} \quad (5.24)$$

an hermitian operator.

One has to introduce this operator in Eq. (5.16) in order to compute the Jacobian. Using (5.16) and (2.10) we have

$$\log J = -2 \int d^2x \int_0^1 dr \operatorname{tr}(K_0(D_r; x, x) \gamma_s) \alpha(x). \quad (5.25)$$

The Kernel K_0 can be evaluated using Seeley's coefficients, as it is sketched in the Appendix. From Eq. (A.9) we have

$$\operatorname{tr}(K_0(D_r; x, x) \gamma_s) = -\frac{e}{2\pi} (1-r) F_{01} \quad (5.26)$$

and hence

$$\log J = \frac{e}{4\pi} \int \varepsilon_{\mu\nu} F_{\mu\nu} \alpha d^2x \quad (5.27)$$

or, integrating by parts and using (5.19),

$$J = \exp \left(-\frac{e^2}{2\pi} \int A_\mu A_\mu d^2x \right). \quad (5.28)$$

We then see that the "photon" mass term, related to the chiral anomaly, is precisely generated by the chiral Jacobian. We can then write for the generating functional in the presence of fermion sources:

$$\begin{aligned} Z(\bar{\eta}, \eta) = & \int \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \mathcal{D}A \exp \left[- \int \left(-\frac{1}{4} F_{\mu\nu}^2 \right. \right. \\ & + \frac{e^2}{2\pi} A_\mu^2 + \bar{\psi}' i \not{\partial} \psi' + \bar{\psi}' e^{\gamma_5} \eta \\ & \left. \left. + \bar{\eta} e^{\gamma_5} \psi' \right) d^2x \right]. \end{aligned} \quad (5.29)$$

Any Green function can be obtained from (5.29) and the complete solution of the massless Schwinger model follows in a very economical way. Of course Fujikawa's regularization approach gives the same answer (since D_r is hermitian) and in this last form the problem has been discussed in Refs. [20, 22].

The Thirring Model [21]

This is a purely fermionic model (for the original works see the references in [21]):

$$\mathcal{L}_{\text{Th}} = \bar{\psi} i \not{\partial} \psi - \frac{1}{2} g^2 (\bar{\psi} \gamma_\mu \psi)^2. \quad (5.30)$$

Now, one can introduce an auxilliary vector field A_μ in the generating functional

$$Z_{\text{Th}} = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int \mathcal{L}_{\text{Th}} d^3x \right) \quad (5.31)$$

through the identity

$$\begin{aligned} & \exp \left\{ \frac{g^2}{2} \int (\bar{\psi} \gamma_\mu \psi)^2 d^2x \right\} \\ &= \int \mathcal{D}A_\mu \exp \left\{ - \int \left[g \bar{\psi} \not{A} \psi + \frac{1}{2} A_\mu A_\mu \right] d^2x \right\}, \end{aligned} \quad (5.32)$$

and hence passing to an effective vector theory

$$Z_{\text{Th}} = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A_\mu \exp \left\{ - \int [\bar{\psi} (i \not{\partial} + g \not{A}) \psi + \frac{1}{2} A_\mu^2] d^2x \right\} \quad (5.33)$$

The decoupling of the fermion fields takes place now if one performs the change of variables

$$\begin{aligned} \psi &= e^{i\eta(x) + \gamma_5 \phi(x)} \chi \\ \bar{\psi} &= \bar{\chi} e^{-i\eta(x) + \gamma_5 \phi(x)} \\ A_\mu &= -\frac{1}{g} \varepsilon_{\mu\nu} \partial_\nu \phi + \frac{1}{g} \partial_\mu \eta. \end{aligned} \quad (5.34)$$

This is analogous to the Schwinger model case except that there is now an additional η -field owing to the fact that in this theory A_μ can have a longitudinal part. The computation of the fermionic Jacobian follows exactly the same steps as before. The final result is

$$J = \exp \left\{ -\frac{1}{\pi} \int d^2x \left[\frac{1}{2} (\partial_\mu \phi)^2 + \varepsilon_{\mu\nu} \partial_\mu \phi \partial_\nu \eta \right] \right\}. \quad (5.35)$$

This leads to an effective Lagrangian

$$\mathcal{L}_{\text{eff.}} = \bar{\chi} i \not{\partial} \chi + \frac{1}{2g^2} \left(1 + \frac{g^2}{\pi} \right) (\partial_\mu \phi)^2 + \frac{1}{2g^2} (\partial_\mu \eta)^2, \quad (5.36)$$

and, from the generating functional

$$Z = \int \mathcal{D}\phi \mathcal{D}\eta \mathcal{D}\bar{\chi} \mathcal{D}\chi \exp \left\{ - \int [\mathcal{L}_{\text{eff.}} + \text{sources}] d^2x \right\}, \quad (5.37)$$

all Green functions can be computed and coincide with the Klaiber solution (see Ref. [21]).

Two-Dimensional QCD [22]

The Lagrangian is

$$\mathcal{L} = \bar{\psi}(i\not{\partial} + g\not{A})\psi + \frac{1}{2} \text{tr} F_{\mu\nu} F_{\mu\nu}, \quad (5.38)$$

where the fermions are taken in the fundamental representation of $SU(N)$ and

$$A_\mu = A_\mu^a t^a \quad (5.39)$$

with t^a being the $N^2 - 1$ generators of $SU(N)$.

It has been proved that [37] in analogy with the Lorentz gauge in the $U(1)$ case, there exists a gauge which we call the decoupling one, in which A_μ can be written in the form

$$A = -\frac{i}{g} \not{\partial}(e^{\gamma_5 \phi}) e^{-\gamma_5 \phi} \quad (5.40)$$

with ϕ taking values in the Lie Algebra of $SU(N)$. In this gauge, which has not been characterized yet in a differential way (see, however, Ref. [37]), the change of variables

$$\begin{aligned} \psi &= e^{\gamma_5 \phi} \chi \\ \bar{\psi} &= \bar{\chi} e^{\gamma_5 \phi} \end{aligned} \quad (5.41)$$

decouples the fermion fields from the gluons, leading, at the classical level, to

$$\mathcal{L} = \bar{\chi} i \not{\partial} \chi + \frac{1}{4} (F_{\mu\nu})^2. \quad (5.42)$$

The Jacobian computation is now made difficult by the non-Abelian character of the model. It has not been computed in a closed form but for the $SU(2)$ case. One can infer, however, that the Schwinger mechanism takes place and the gluons get a mass, in analogy with the $U(1)$ case. Just for the sake of completeness we give the result for the $SU(2)$ Jacobian [37]:

$$\begin{aligned} J = \exp \left\{ -\frac{1}{2\pi} \int d^2x \varepsilon_{\mu\nu} \phi \cdot (\partial_\mu \mathbf{A}_\nu - \partial_\nu \mathbf{A}_\mu) \right. \\ \left. + \mathbf{A}_\mu \times \mathbf{A}_\nu \left(\frac{1}{\phi \tanh \phi} - \frac{1}{\text{Sh}^2 \phi} \right) \right\}, \end{aligned} \quad (5.43)$$

where $\phi = \phi \cdot \boldsymbol{\tau}$ and $A_\mu = \mathbf{A}_\mu \cdot \boldsymbol{\tau}$. As stressed in Ref. [22], by inspection of the resulting effective Jacobian one can confirm the arguments presented in Ref. [38] on the properties of this model.

The Chiral Gross–Neveu Model [21]

The Lagrangian of this model can be written, after a Fierz-transformation, as

$$\mathcal{L} = \bar{\psi} i \not{\partial} \psi - \frac{g^2}{N} (\bar{\psi} \gamma_\mu \lambda^a \psi)^2, \quad (5.44)$$

where λ^a are the generators of the $U(N)$ group. The fermions are taken in the fundamental representation of $SU(N)$ and the model is invariant under $U(1) \times U(1)$ and $SU(N)$ (both global) transformations. Using the non-Abelian version of identity (5.32) one can introduce N^2 vector fields A_μ taking values in the Lie algebra of $U(N)$,

$$A_\mu = A_\mu^0 \frac{I}{2} + A_\mu^a t^a, \quad (5.45)$$

with t^a being the $N^2 - 1$ generators of $SU(N)$:

$$\begin{aligned} & \exp \left\{ \frac{g^2}{2N} \int d^2x (\bar{\psi} \gamma_\mu \lambda^a \psi)^2 \right\} \\ &= \int \mathcal{D}A_\mu^0 \mathcal{D}A_\mu^a \exp \left\{ - \int d^2x \left[\frac{g}{\sqrt{N}} \bar{\psi} \left(A^a t^a + A^0 \frac{I}{2} \right) \psi \right. \right. \\ & \quad \left. \left. + \frac{1}{2} A_\mu^a A_\mu^a + \frac{1}{2} A_\mu^0 A_\mu^0 \right] \right\}. \end{aligned} \quad (5.46)$$

The decoupling change of variables is just a combination of the Thirring and QCD_2 cases

$$\begin{aligned} \psi &= e^{i\eta_0 + \gamma_5 \phi_0} u^{-1} u_5 \chi \\ \bar{\psi} &= \bar{\chi} u_5 u e^{-i\eta_0 + \gamma_5 \phi_0} \\ A_\mu^0 &= -\frac{\sqrt{N}}{g} (\epsilon_{\mu\nu} \partial_\nu \phi_0 - \partial_\mu \eta_0) \\ A_\mu &= u^{-1} \left(\frac{i\sqrt{N}}{g} \partial(u_5) \right) u_5^{-1} u - u^{-1} i \frac{\sqrt{N}}{g} \partial(u), \\ u &= e^{it \cdot \eta(x)}, \quad u_5 = e^{\gamma_5 t \cdot \phi(x)}. \end{aligned} \quad (5.47)$$

Again, the computation of the Jacobian follows the steps described before. In spite of the difficulties originated by the non-Abelian $SU(N)$ sector, important features can be discovered, in particular the factorization of the $U(1)$ chiral massless part and the almost long-range order predicted by Witten by means of the $1/N$ expansion [39] (for details, see Ref. [21]).

CPⁿ Models with Fermions [23]

The (Euclidean) Lagrangian is

$$\mathcal{L} = \left| \left(\partial_\mu - \frac{f}{n} (\bar{Z} \vec{\partial}_\mu Z) \right) Z \right|^2 + \bar{\psi} \left(i \not{\partial} - \frac{ief}{n} (\bar{Z} \vec{\partial} Z) \right) \psi. \quad (5.48)$$

Since $\bar{Z} \vec{\partial}_\mu Z$ acts like a gauge field, it can be eliminated from the covariant derivative

by introducing first an auxiliary vector field and then making a chiral rotation: In a first step one gets the effective Lagrangian

$$\begin{aligned}\mathcal{L}_{\text{eff}} &= \overline{D_\mu Z} D_\mu Z + \bar{\psi} \left(i\not{\partial} - \frac{e}{\sqrt{n}} \chi \right) \psi + \frac{fe^2}{2n} (\bar{\psi} \gamma_\mu \psi)^2 \\ D_\mu &= \partial_\mu + \frac{i}{\sqrt{n}} \lambda_\mu,\end{aligned}\tag{5.49}$$

where the auxiliary gauge field λ_μ replaces the quartic term in (5.48). Now, the chiral rotation

$$\begin{aligned}\psi &= e^{\gamma_5(e/\sqrt{n})\Lambda(x)} \chi \\ \bar{\psi} &= \bar{\chi} e^{\gamma_5(e/\sqrt{n})\Lambda(x)} \\ \lambda_\mu &= \varepsilon_{\mu\nu} \partial_\nu A\end{aligned}\tag{5.50}$$

decouples bosons from fermions. At the classical level we have

$$\mathcal{L}_{\text{eff}} = \overline{D_\mu Z} D_\mu Z + \bar{\chi} i\not{\partial} \chi + \frac{fe^2}{2n} (\bar{\chi} \gamma_\mu \chi)^2.\tag{5.51}$$

The Jacobian gives now the contribution

$$J = \exp \left(\frac{-e^2}{2\pi} \int (\partial_\mu A)^2 d^2x \right).\tag{5.52}$$

Combining (5.51)–(5.52) in the generating functional of the theory we discover that the effective Lagrangian $\mathcal{L}_{\text{eff}} + \log J$ has a Thirring-like fermionic part decoupled from the purely bosonic massive sector which can be treated by using a $1/n$ expansion (see the references in [23]). Then the analysis of the model can be very easily done and one discovers the factorization of a $U(1)$ chiral factor in the Green functions and a screening phenomenon.

The θ -Vacuum [32]

Consider the generating functional for a gauge theory with massless fermions when the θ -vacuum term [7–9] is considered

$$Z[\theta] = \int \mathcal{D}A_\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left\{ - \int d^d x [\bar{\psi} \not{D} \psi + \tfrac{1}{4} F_{\mu\nu}^2 + \theta \rho_d] \right\},\tag{5.53}$$

$$\rho_d = \begin{cases} -\frac{e}{4\pi} \varepsilon_{\mu\nu} F_{\mu\nu} & \text{for } d=2, U(1) \text{ gauge theory} \\ -\frac{e^2}{16\pi^2} \text{tr} {}^*F_{\mu\nu} F_{\mu\nu} & \text{for } d=4, SU(N) \text{ gauge theory.} \end{cases}\tag{5.54}$$

It is in general considered that a finite global chiral rotation can eliminate any reference to θ owing to the anomalous chiral symmetry. In the path-integral framework, this can be seen by making the rotation (5.1) with angle $\alpha = -\theta/2$. since the corresponding Jacobian [32], Eq. (3.16),

$$J = \exp \left\{ \theta \int d^d x \rho_d \right\} \quad (5.55)$$

exactly cancels the θ -term in (5.53). Hence

$$Z[\theta] = Z[0]. \quad (5.56)$$

We know, however, that the theory is not θ -independent. As was stressed by Wilczek [40], the puzzle can be solved by noting that quantities involving chiral non-singlet operators keep trace of the θ -term. Suppose we want to compute the vacuum expectation value

$$W_\theta(x) = \langle \bar{\psi}(x) \psi(x) \bar{\psi}(0) \psi(0) \rangle_\theta = \frac{\delta^2 Z[\theta, j, j]}{\delta j(x) \delta j(0)} \Big|_{j=0}, \quad (5.57)$$

where we have defined

$$Z[\theta, j, k] = \int \mathcal{D}A \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left\{ - \int d^d x \left[\mathcal{L} + \theta \rho_d + j(x) \bar{\psi} \frac{(1 + \gamma_5)}{2} \psi + k(x) \bar{\psi} \frac{(1 - \gamma_5)}{2} \psi \right] \right\}. \quad (5.58)$$

If we now perform in $Z[\theta, j, k]$ the chiral rotation, the Jacobian will be different from (5.55) since we know that the complete Dirac operator appearing in (5.58) will determine the Jacobian form. For $\alpha = -\theta/2$ the result is (Eqs. (3.31a) and (3.31b))

$$\begin{aligned} J_2 &= \exp \left(-2\alpha \int d^2 x \rho_2 \right) \exp \left\{ \frac{1}{8\pi} \int d^2 x [j^2(e^{4\alpha} - 1) + k^2(e^{-4\alpha} - 1)] \right\}, \\ J_4 &= \exp \left(-2\alpha \int d^4 x \rho_4 \right) \exp \left\{ -\frac{1}{8\pi^2} \int d^4 x \left[\frac{1}{24} (j^4(e^{8\alpha} - 1) + k^4(e^{-8\alpha} - 1)) \right. \right. \\ &\quad \left. \left. - \frac{1}{3} jk(j^2(e^{4\alpha} - 1) + k^2(e^{-4\alpha} - 1)) \right. \right. \\ &\quad \left. \left. + \frac{1}{12} (k\partial^2 k(1 - e^{-4\alpha}) + j\partial^2 j(1 - e^{4\alpha})) \right] \right\}. \end{aligned} \quad (5.59)$$

The presence of sources therefore makes it impossible to completely eliminate the θ -

dependence. Additional δ -terms appear with θ -factors. For example, for the $d = 2$ case, we have (Fourier transforming and continuing to Minkowski space)

$$W_\theta(q) = W_{\theta=0}(q) + \frac{1}{(2\pi)^3} (\cos 2\theta - 1). \quad (5.60)$$

Inspection of (5.59) shows that δ -terms (and its derivatives) appear for $d = 4$ when differentiating Z . The contributions are analogous to the Schwinger terms arising from anomalous commutators [3, 41].

6. SUMMARY AND CONCLUSIONS

We have discussed in this work the change in the fermionic path-integral measure under chiral rotations, motivated by the interest in the study of chiral symmetries without resort to diagrammatic analysis.

Using the ζ -function method, we have been able to extend in a natural way the evaluation of chiral Jacobians to theories including non-hermitian Dirac operators, this being the case of models including γ_5 -couplings (e.g., in the Weinberg-Salam case) or related to the relativistic string [42].

In our approach, there is no necessity of constructing an hermitian operator from D in order to regularize the Jacobian. Therefore, no choice between "complete," "incomplete" or analytically extended Dirac operator appears (as it happens in other approaches). Although we prove in Section 4 that the different methods are equivalent for hermitian, the explicit calculations of Sections 2 and 3 show that for non-hermitian operators the results disagree.

The ζ -function approach we developed seems very natural in the sense that it is based on the definition of the quadratic fermionic path-integral as a regularized determinant which is defined in terms of the ζ -function independently of the hermiticity of D . It will be on physical grounds, however, that the choice of regularization prescription in the different models has to be done.

We have also shown how finite chiral transformations can be handled by iteration, at least for the cases one does not take the fermions from one topological sector to another. In spite of this restriction we have been able to obtain a complete and simple solution of the Schwinger or Thirring models as well as new insights on other interesting two-dimensional models. Finally, we mention that a careful treatment of chiral rotations allowed us to study some features of gauge theories related to the θ -vacuum.

APPENDIX

We shall describe in this Appendix how to evaluate the Kernel $K_0(D; x, x)$ in terms of Seeley's coefficients [31].

For the sake of simplicity, we are going to show in detail the computation for the simple $d=2$, $U(1)$ case and then briefly sketch the more interesting $d=4$ case discussed in Section 2.

As we explained in Section 5, in order to solve the Schwinger model in the path-integral framework, one is faced with the evaluation of a chiral Jacobian (Eq. (5.21))

$$\log J = -2 \iint_0^1 \text{tr}(K_0(D_r; x, x) \gamma_s) a(x) dx dr, \quad (\text{A.1})$$

where tr means the matricial trace and $K_0(D_r; x, x)$ is the Kernel of the operator D_r^{-s} analytically extended to $s=0$. The operator D_r is given by expression (5.4)

$$D_r = i\not{\partial} + e(1-r)\not{A}, \quad 0 \leq r \leq 1. \quad (\text{A.2})$$

The symbol of D_r is

$$\sigma(D_r) = a_1 + a_0, \quad (\text{A.3})$$

where

$$\begin{aligned} a_1 &= -\not{\xi} \\ a_0 &= e(1-r)\not{A} = \not{B}. \end{aligned} \quad (\text{A.4})$$

Following Seeley [31] we have to construct the coefficients

$$\begin{aligned} b_{-1}(x, \xi, \lambda) &= (a_1 - \lambda)^{-1} \\ b_{-2}(x, \xi, \lambda) &= -b_{-1}a_0b_{-1} \\ b_{-3}(x, \xi, \lambda) &= \frac{\partial b_{-1}}{\partial \xi_\mu} i \frac{\partial a_0}{\partial x_\mu} b_{-1} - b_{-2}a_0b_{-1} \end{aligned} \quad (\text{A.5})$$

and then use the expression

$$K_0(D_r; x, x) = \frac{-i}{(2\pi)^2} \int_{|k|=1} \int_0^\infty b_{-3}(x, \xi, iu) du d\xi. \quad (\text{A.6})$$

From (A.4) and (A.5) we find

$$\begin{aligned} b_{-1} &= \frac{1}{(\lambda^2 - \xi^2)} (\not{\xi} - \lambda I) \\ b_{-2} &= \frac{1}{(\lambda^2 - \xi^2)^2} (2\lambda \xi_\mu B_\mu I - (\lambda^2 - \xi^2) \not{B} - 2\xi_\mu B_\mu \not{\xi}) \\ b_{-3} &= \frac{i}{(\lambda^2 - \xi^2)^3} [(2(\not{\xi} - \lambda I) \xi_\mu \partial_\mu + (\lambda^2 - \xi^2) \not{\partial}) \not{B} (\not{\xi} - \lambda I) \\ &\quad - i(2\lambda \xi_\mu B_\mu I - (\lambda^2 - \xi^2) \not{B} - 2\xi_\mu B_\mu \not{\xi})(\lambda \not{B} - \not{B} \not{\xi})]. \end{aligned} \quad (\text{A.7})$$

Taking into account the γ -matrices properties we get

$$\text{tr}(b_{-3}\gamma_5) = \frac{-i\lambda}{(\lambda^2 - \xi^2)^2} \text{tr}(\not{\partial}\not{\partial}\gamma_5) = -2e(1-r) \frac{\lambda}{(\lambda^2 - \xi^2)^2} F_{01}(x), \quad (\text{A.8})$$

and using (A.8) and (A.6)

$$\text{tr}(K_0(D_r; x, x) \gamma_5) = \frac{-e}{2\pi} (1-r) F_{01} \quad (\text{A.9})$$

which is the result stated in Eq. (5.26).

For the $d=4$ case, with D given by Eq. (2.12)

$$D = i\not{\partial} + \gamma_5 \not{A} \quad (\text{A.10})$$

we have to compute (Eq. (2.16))

$$\text{tr}(\gamma_5 K_0(D; x, x)) = \frac{-i}{(2\pi)^4} \int_{|\xi|=1} \int_0^{\infty} \text{tr}(\gamma_5 b_{-5}(x, \xi, iu)) du d\xi. \quad (\text{A.11})$$

According to Eq. (2.17) the b 's are given by

$$\begin{aligned} b_{-1}(x, \xi, \lambda) &= (a_1 - \lambda I)^{-1} \\ b_{-2}(x, \xi, \lambda) &= -b_{-1} a_0 b_{-1} \\ b_{-3}(x, \xi, \lambda) &= -b_{-2} a_0 b_{-1} - \frac{\partial b_{-1}}{\partial \xi_\mu} i \frac{\partial a_0}{\partial x_\mu} b_{-1} \\ b_{-4}(x, \xi, \lambda) &= -b_{-3} a_0 b_{-1} - \frac{\partial b_{-2}}{\partial \xi_\mu} i \frac{\partial a_0}{\partial x_\mu} b_{-1} \\ &\quad + \frac{1}{2} \frac{\partial^2 b_{-1}}{\partial \xi_\mu \partial \xi_r} \frac{\partial^2 a_0}{\partial x_\mu \partial x_r} b_{-1} \\ b_{-5}(x, \xi, \lambda) &= -b_{-4} a_0 b_{-1} - \frac{\partial b_{-3}}{\partial \xi_\mu} i \frac{\partial a_0}{\partial x_\mu} b_{-1} \\ &\quad + \frac{1}{2} \frac{\partial^2 b_{-2}}{\partial \xi_\mu \partial \xi_r} \frac{\partial^2 a_0}{\partial x_\mu \partial x_r} b_{-1} + \frac{i}{6} \frac{\partial^3 b_{-1}}{\partial \xi_\mu \partial \xi_r \partial \xi_\rho} \frac{\partial^3 a_0}{\partial x_\mu \partial x_r \partial x_\rho} b_{-1}. \end{aligned} \quad (\text{A.12})$$

After a tedious but straightforward computation one gets

$$\text{tr}(K_0(D; x, x) \gamma_5) = -\frac{1}{16\pi^2} \frac{1}{3} \text{tr} {}^*F_{\mu\nu} F_{\mu\nu} - \frac{i}{8\pi^2} \frac{1}{3} \text{tr} \partial_\mu (-4A_\alpha A_\alpha A_\mu + \partial^2 A_\mu) \quad (\text{A.13})$$

which is the result stated in Eq. (2.18).

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